



ECONOMIC MODELS IN ASIA

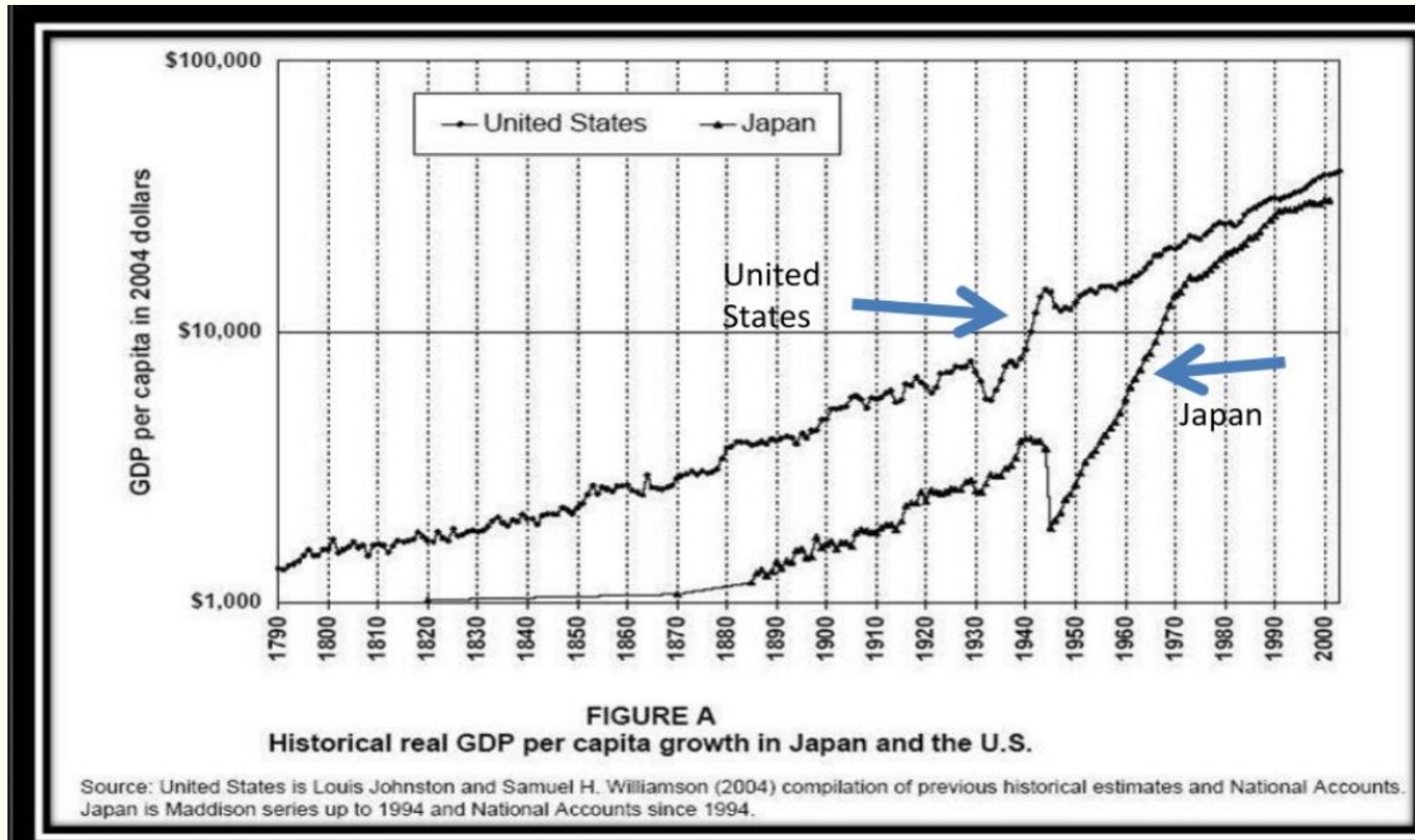
The Rise and Stagnation of Japan: an Exception?



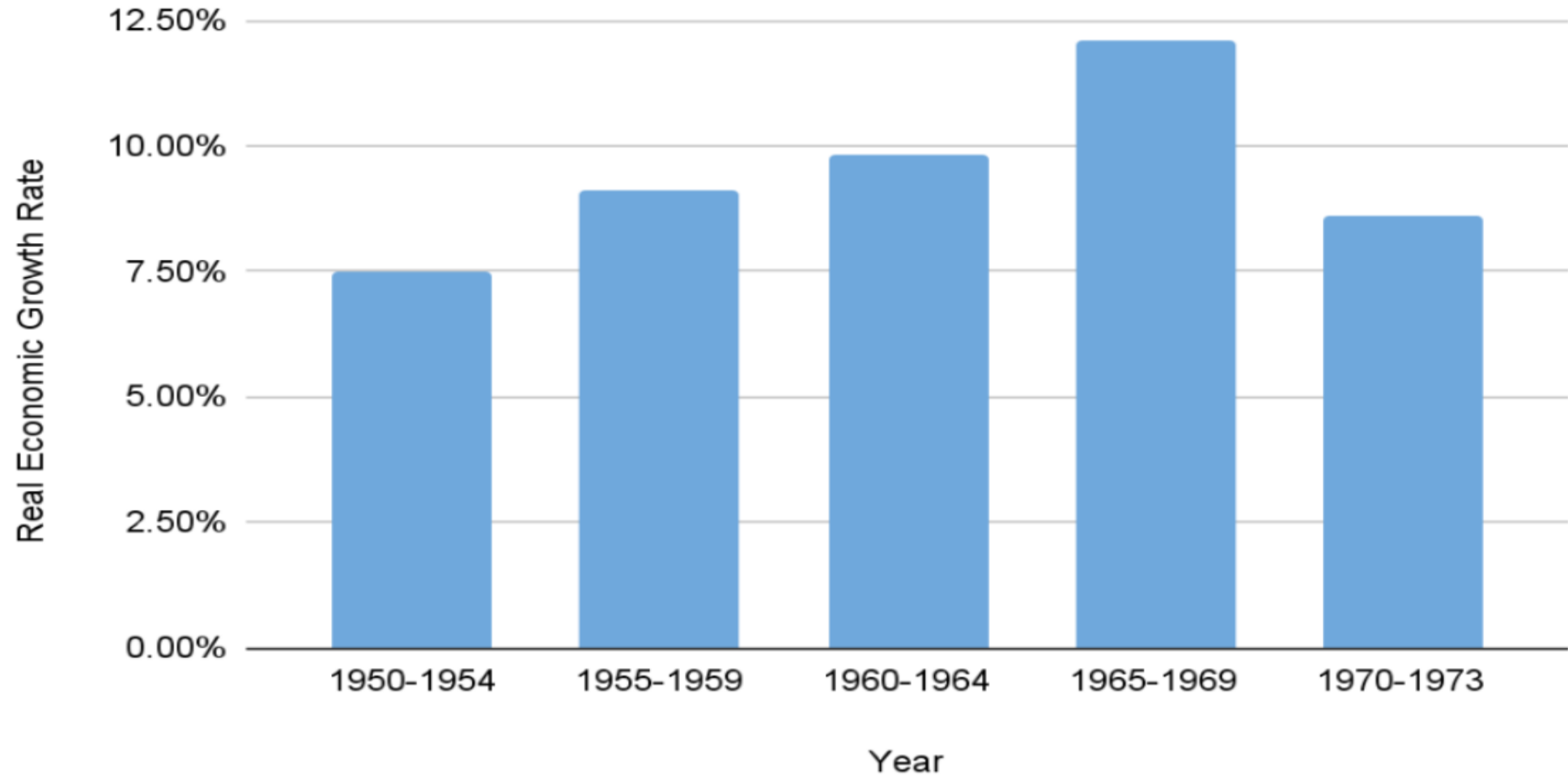
Japan Economic Miracle after the End of WWII

- Japan was devastated at the end of the war: a cost estimated to \$56 Billion and around 3 millions lives
- Sustained economic growth from 1945 to 1991, particularly strong during the first 20 years (see real economic growth)
- 4 main factors explain this «miracle»:
 - Technological change
 - Accumulation of capital
 - Increased quantity and quality of labor
 - Increased international trade
- A combination of strategic planning and firms' cooperation

Historical Real GDP per Capita – Japan and USA



Japan Real Economic Growth 1950-1973



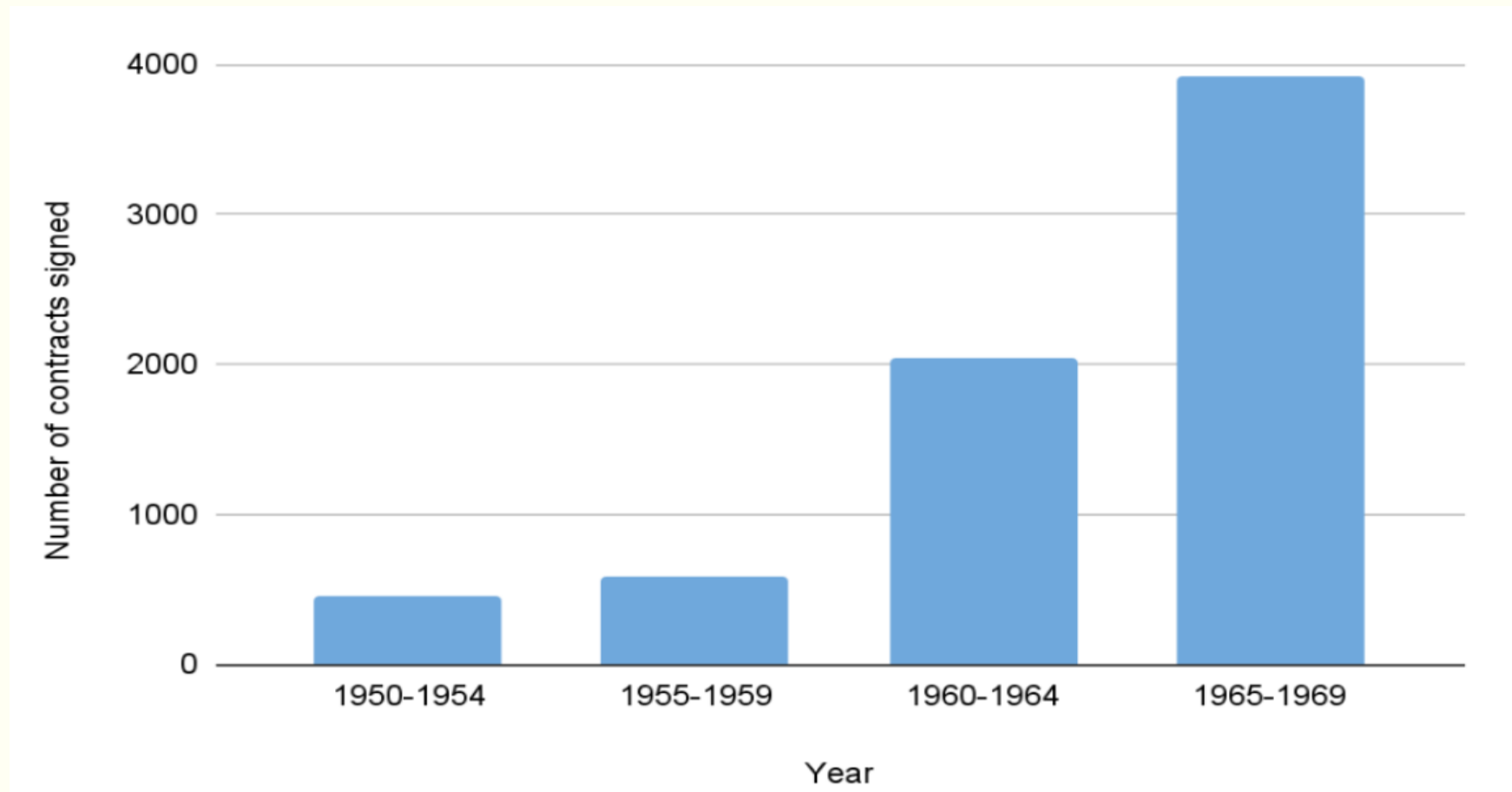
Technicological Change

- After the War, Japan had lost $\frac{1}{4}$ of Industrial capacity + Depreciated stock
- Condition for Japan to adopt an abundance of new technologies without having to wait for assets fully depreciated
- Favorable to innovation
- Fuel innovation
- Accelerated by import of new technology in industries such as iron, steel, petrochemicals electronics, motor manufacturing and chemical engineering

Japan's Technological Change and Productivity

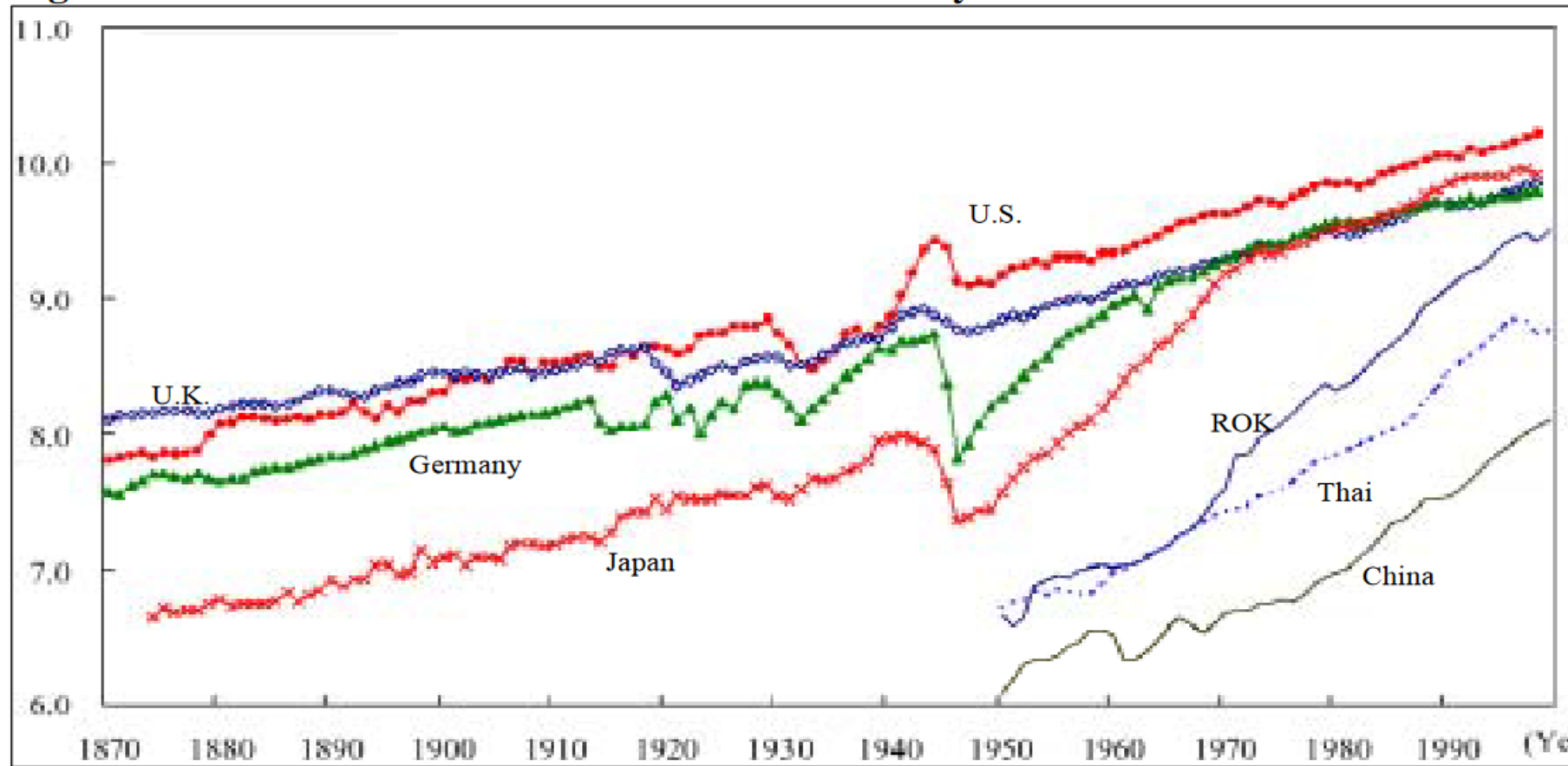
Years	Real GDP	Population	Labor participation	Productivity
1956–65	8.6	0.96	0.36	7.28
1966–75	7.45	1.3	−0.24	6.39
1976–85	3.8	0.78	0.35	2.67
1986–95	2.98	0.37	0.75	1.86
1996–2005	1.13	0.17	−0.20	1.14
2006–15	1.06	−0.11	0.03	1.16

Number of contracts (> 1 year) signed to import new technology in foreign currency between 1950 and 1969.



International Labor Productivity

Figure 2: International Trends in Labor Productivity

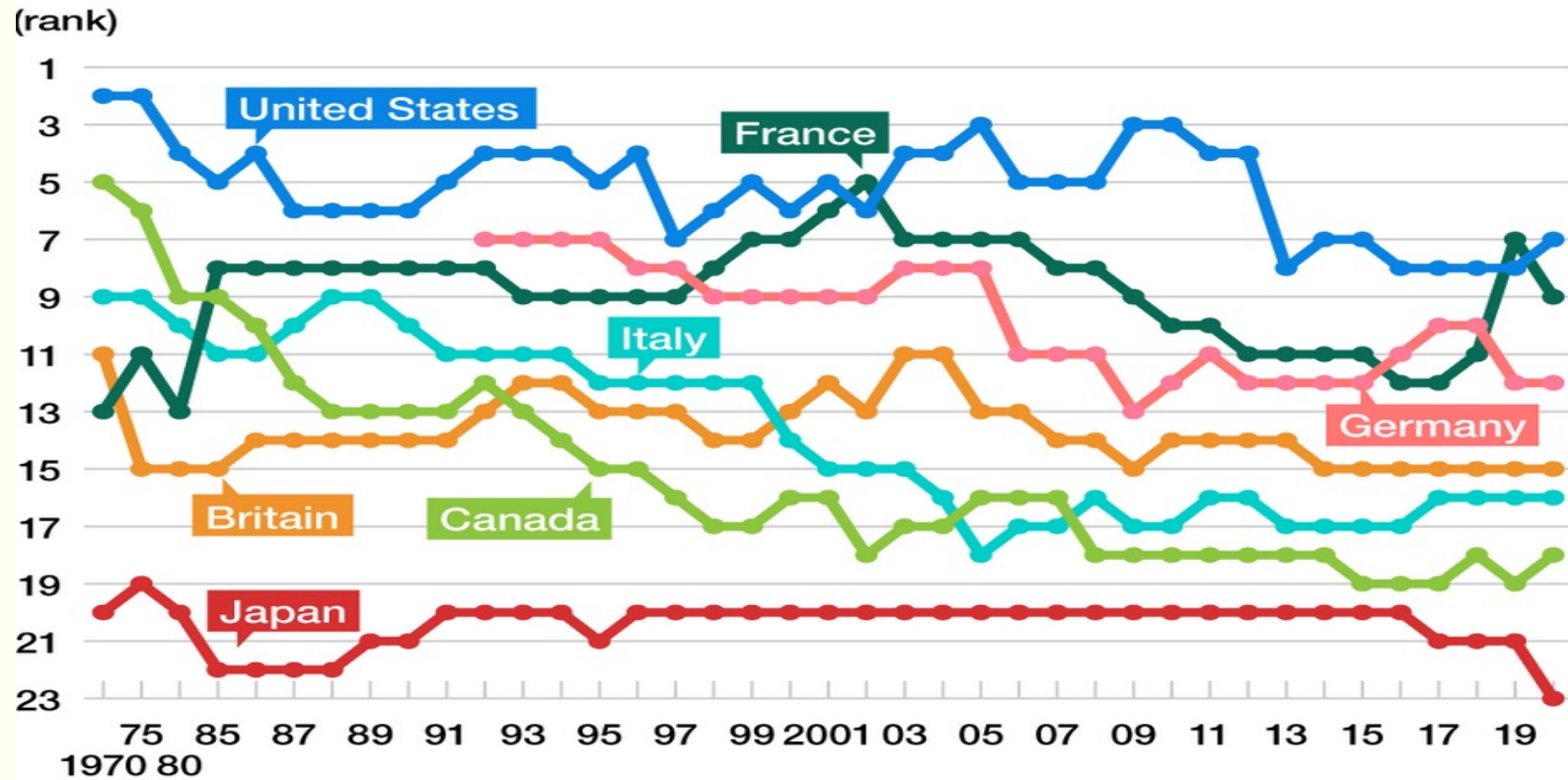


Note: Note: Natural logarithm value of per capita GDP expressed in dollars.

Source: METI (2002): White Paper on International Trade. Data from Maddison, Angus (1995) and (2001).

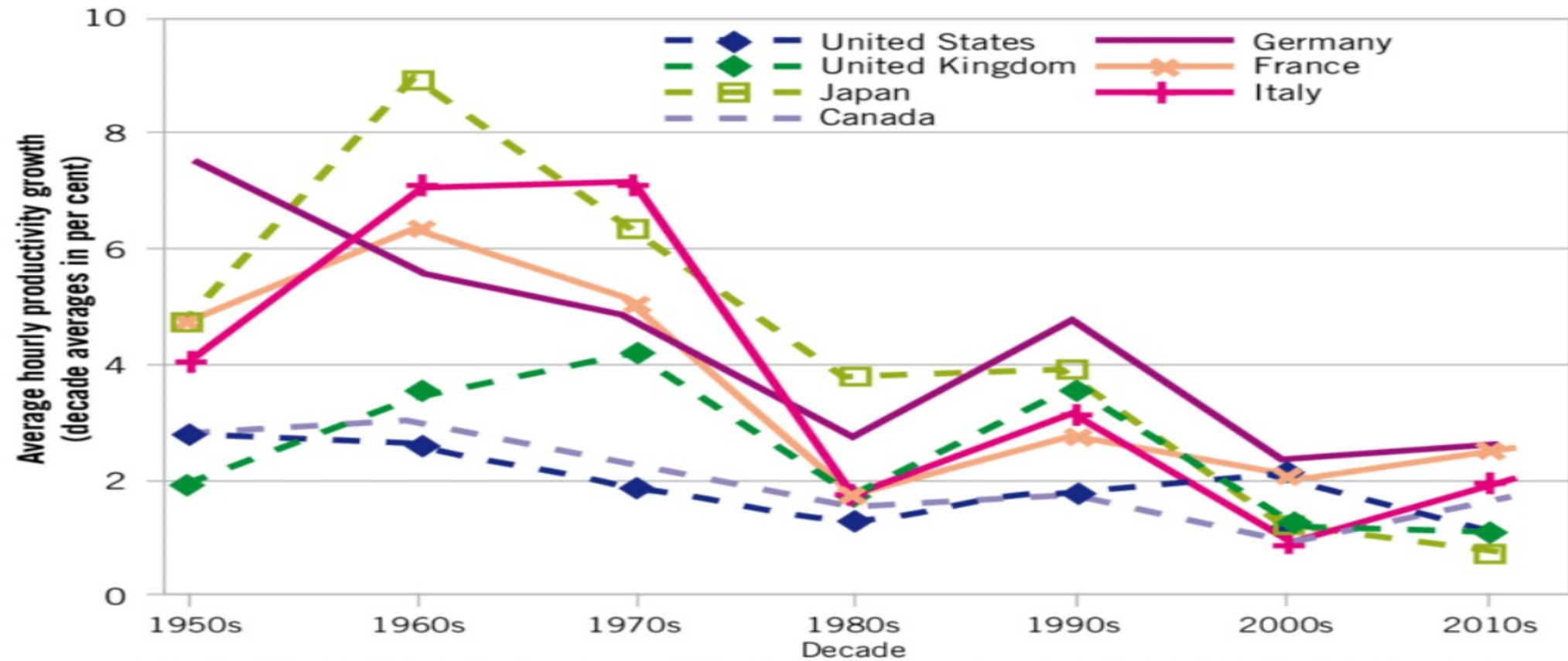
G7 Nations Per-Hour Labor Productivity

Ranking of G7 Nations for Per-Hour Labor Productivity



Created by *Nippon.com* based on data from the Japan Productivity Center.

G-7 Average Hourly Productivity Growth



Source: Penn World Tables version 9, available at: <https://www.rug.nl/ggdc/productivity/pwt/>; authors' calculations.

The Magic Touch of Loose Monetary Policy Combined with Loose Fiscal Policy

- This technological race was fueled by cheap money
- Central bank monetary policy essentially setting the interest rate low.
- Firms were overborrowing and banks were overlending
- Government use generous depreciation provisions and tax deductions:
 - Rapid growing corporations lower tax rate compare to slower-growing firms
- Personal Income Tax: partial exemption of interest and dividend incomes from taxation.

Transfers of Technology

- By importing technology, it fosters innovation:
 - In 1963 the Japan Information Center for Science and Technology wrote 210 000 abstracts of foreign scientific papers
- New technologies imported were improved:
 - Machine tools and robot were imported and used in the automobile industry
 - Imported generators adapted to improve efficiency of electric utility industry
- Technology = 20% more efficient
- 9500 large firms = spending 1/3 of their R&D expenditure and improving imported technique

Capital Accumulation

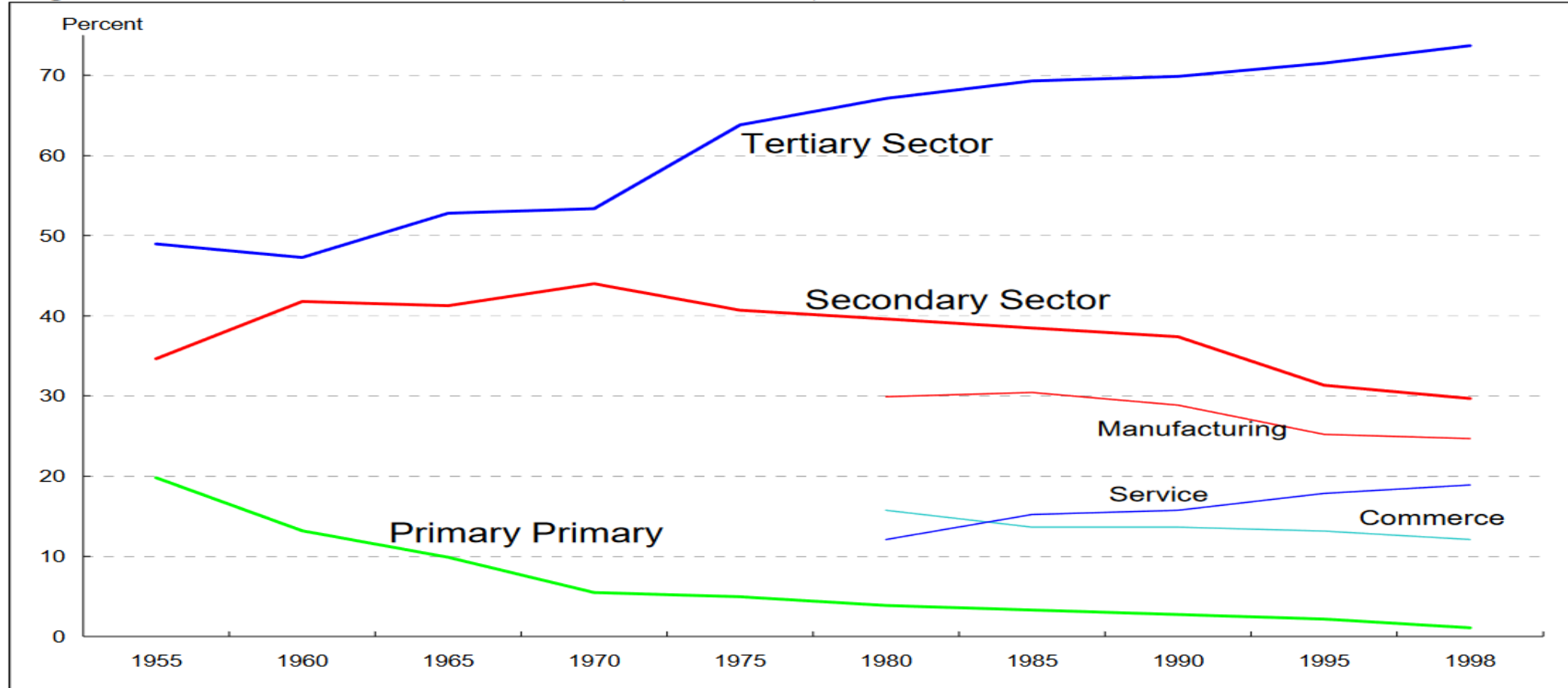
- Concentration of capital in rapid growing industries:
 - In contrast with other countries investing in housing or social overhead capital
- Strong commitment to close the technological gap in order to increase international competitiveness
- High return on investment because of the high marginal productivity
- High rate of savings have avoided high inflation:
 - 1950-1970: personal savings rate = 18,5% Japan 7% US

Quantity and Quality of Labor

- NBER paper finds that the increase in quantity and quality of labor = 30% of the Japanese economic growth
- After the war, large increase in labor so wages increased less rapidly than productivity in the 1950's
- In the 1960's, wage increase in line with productivity
- Labor moved from low productivity sector like agriculture to high productivity (automobile, electronics, aviation...)

Postwar Sector Trends

Figure 5: Postwar Sector Trends (% of GDP)



Note: Percent of nominal GDP. Data from 1955-1985 are based on SNA68; from 1990-98 on SNA93.

The Role of Kereitsu

- Kereitsu = large business groups linking banks, trading companies industrialist through ownership or stock / long standing exclusive relationship
- Size gives financial strength and connections
- Kereitsu outperform national and international competitors in high growth sector
- Government protects Kereitsu
- Ministry of industrial trade and Industry allowing administrative cartels + collusive activities:
 - Mergers amongst 5 largest firms in a given industry
- Japanese business environment very competitive

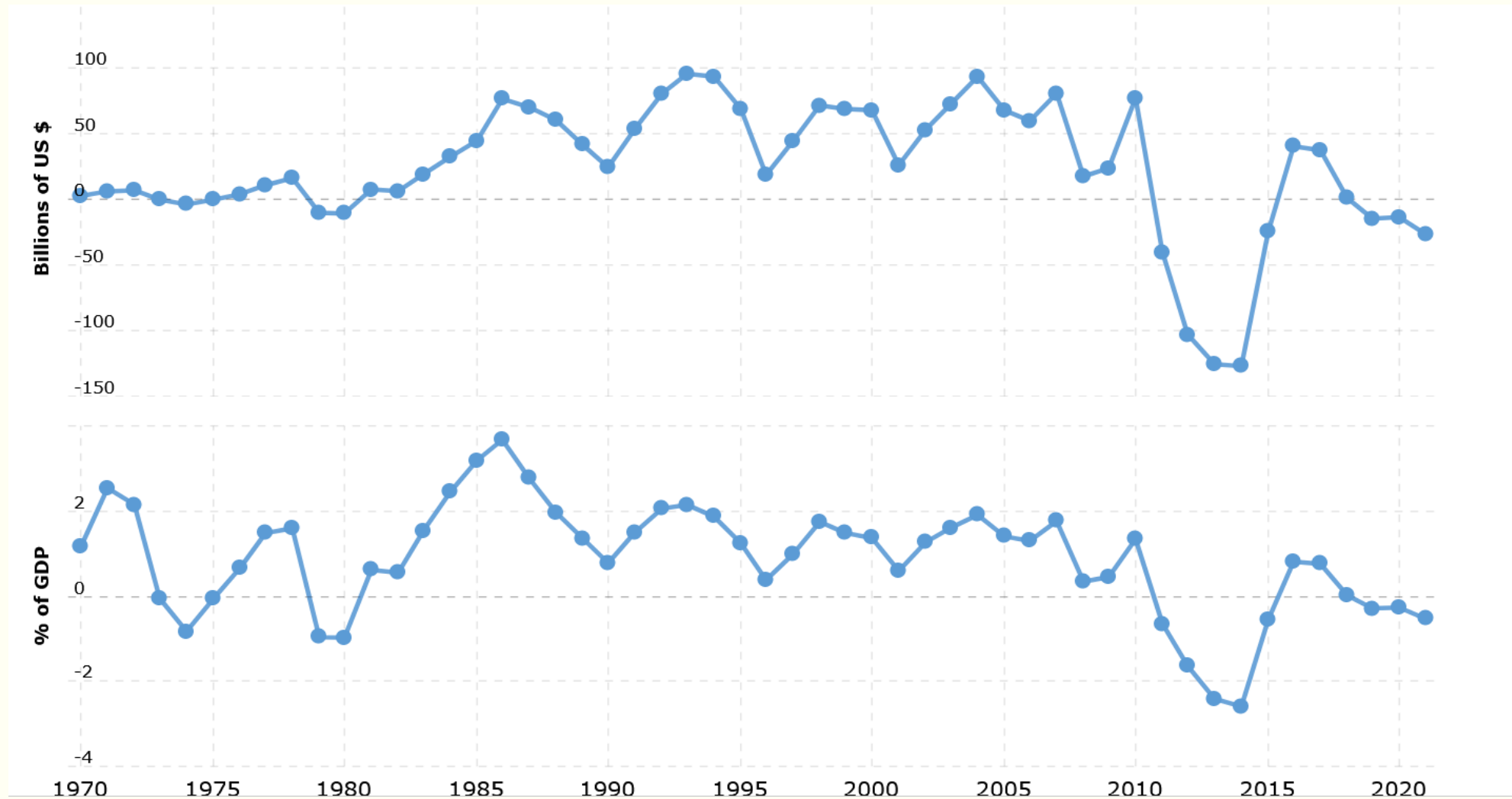
The Role of Kereitsu

- To keep up with product design and implement new production techniques, employees work long hours and are loyal to the firm
- High standards with strict rules
- Management practices innovation:
 - Total quality control
 - Lean production
 - Cross-functional product teams
 - Just in time

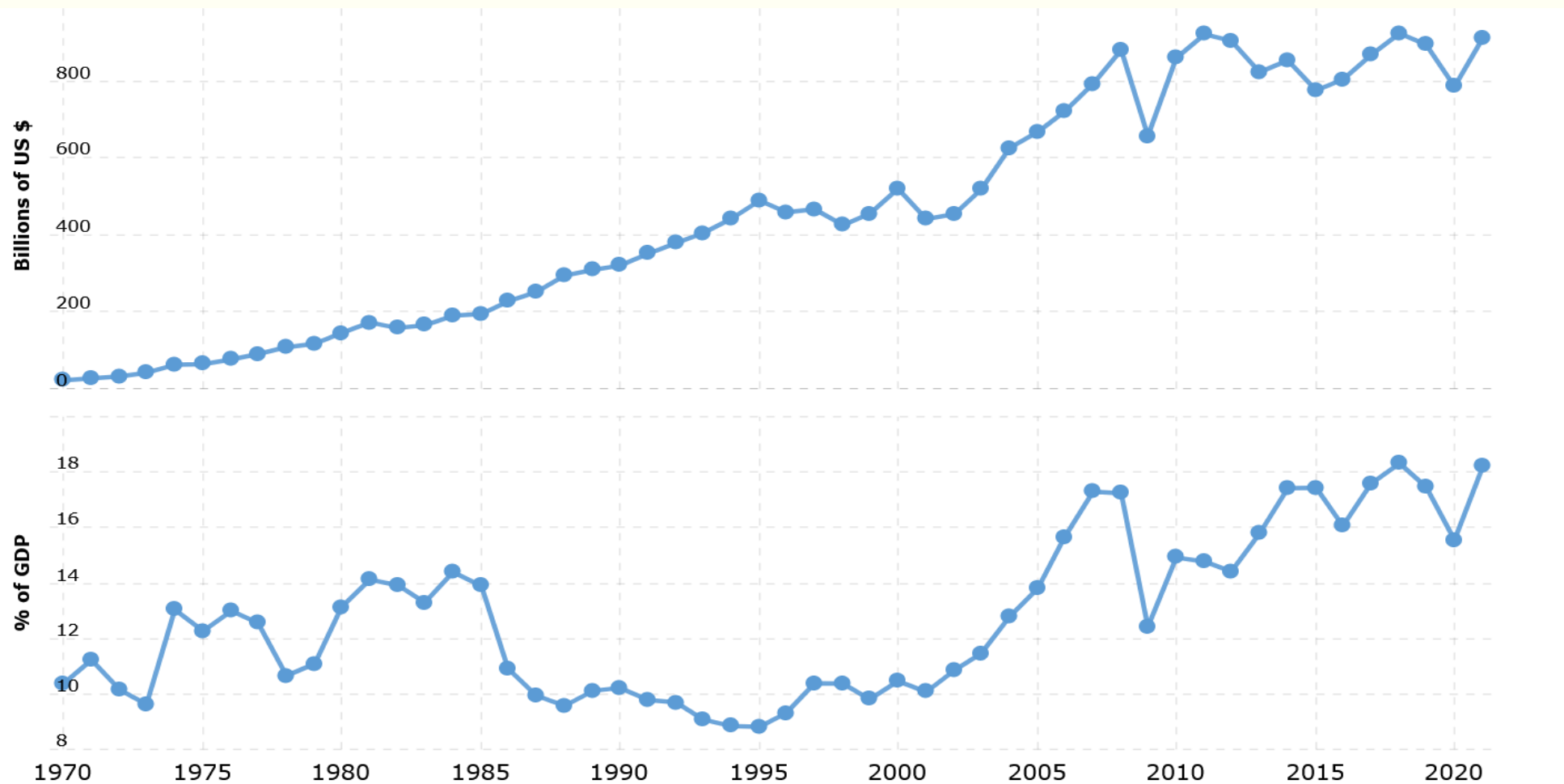
International Trade

- Exports increase rapidly after the war:
 - Tax deductions for overseas sales expenditures + preferential loans = competitive exports
- Mergers + Anti-competitive behavior encouraged in export sectors
- Key factor to stay ahead of international competitor = ability to change exports
 - 1950-1965 = textile and sundry goods to machinery and then to metals
- Thanks to increased efficiency + corporations' ability = ability to stay competitive
- Ability to produce goods most demanded

Japan Trade Balance 1970-2024



Japan Exports 1970-2024



Private Demand and Industrial Sector Shares in the US

Figure 7: Private Demand and Industrial Sector Shares in the U.S.

